

Final Exam — Chemistry

Part A: Multiple Choice

Use your reference table and your knowledge of chemistry to answer the questions below. [35]

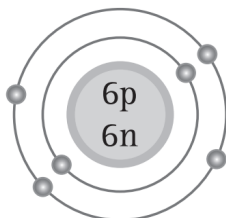
1. The atomic number of calcium is 20. How many valence electrons does a calcium atom have?

a. 1 c. 2
b. 8 d. 20

2. What isotope is produced by the alpha decay of a nucleus of gadolinium-148?

a. samarium-144
b. europium-148
c. terbium-149
d. dysprosium-152

3. What element is represented by this shell model?



a. aluminum c. carbon
b. oxygen d. fluorine

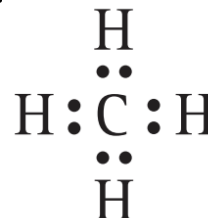
4. What type of bonding holds the atoms of ethanol, C_2H_6O , together?

a. ionic
b. metallic
c. nuclear
d. covalent

5. Which family of metals on the periodic table is the most reactive overall?

a. halogens
b. Alkaline earth metals
c. Alkali metals
d. Noble gases

6. What is the shape of the methane, CH_4 molecule shown below?



a. tetrahedral c. linear
b. bent d. pyramidal

7. A 5.00 L volume of air is cooled from 200. K to 100. K. What is the volume of the heated air if the pressure remains constant?

a. 1.00 L c. 2.50 L
b. 10.0 L d. 5.00 L

8. Convert 315 K to $^{\circ}C$.

a. $42^{\circ}C$ c. $283^{\circ}C$
b. $347^{\circ}C$ d. $588^{\circ}C$

9. This table shows the pressure, volume, and temperature of a gas under different conditions. What is the missing value on the data table?

Pressure	Volume	Temperature
2.0 atm	8.0 L	$300^{\circ}C$
2.0 atm	4.0 L	$14^{\circ}C$
4.0 atm	4.0L	$300^{\circ}C$
4.0 atm	?	$14^{\circ}C$

a. 1.0 L c. 2.0 L
b. 4.0 L d. 8.0 L

10. How many moles of argon are there in a container that holds 44.8 L, 273 K and 5.0 atm pressure when $r = 0.08206 \text{ L}\cdot\text{atm}/\text{mol}\cdot\text{K}$

- a. 2
- b. 4
- c. 6
- d. 10.

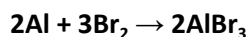
11. Which chemical equation is correctly balanced?

- a. $\text{Al}(\text{OH})_3 \rightarrow \text{Al}_2\text{O}_3 + 2\text{H}_2\text{O}$
- b. $\text{KI} + \text{Pb}(\text{NO}_3)_2 \rightarrow \text{KNO}_3 + \text{PbI}$
- c. $2\text{Na}_3\text{PO}_4 + 3\text{CaCl}_2 \rightarrow 6\text{NaCl} + \text{Ca}_3(\text{PO}_4)_2$
- d. $\text{C}_5\text{H}_{12} + 6\text{O}_2 \rightarrow 5\text{CO}_2 + 6\text{H}_2\text{O}$

12. Based on their chemical structures, which of these materials is likely to be soluble in water?

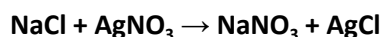
- a. CCl_4
- b. CH_4
- c. HCl
- d. O_2

13. According to this balanced equation, how many moles of bromine are required to produce 15 mol of aluminum bromide?



- a. 20 mol
- b. 22.5 mol
- c. 30 mol
- d. 45 mol

14. How is this chemical reaction classified?



- a. combustion
- b. synthesis
- c. single replacement
- d. double replacement

15. Which solution of sodium bromide in water has the highest molarity?

Solution	NaBr (g)	Volume (mL)
A	78	1000
B	53	500
C	32	300
D	30	250

- a. Solution A
- b. Solution B
- c. Solution C
- d. Solution D

16. What is the molar mass of $\text{Cu}(\text{NO}_3)_2$?

- a. 93.5521 g/mol
- b. 187.5558 g/mol
- c. 125.5509 g
- d. 187.5558 g

17. If you have 3 mol of glucose in 6 L of solution, what is the concentration of the solution?

- a. 0.5 M
- b. 2 M
- c. 4 M
- d. 8 M

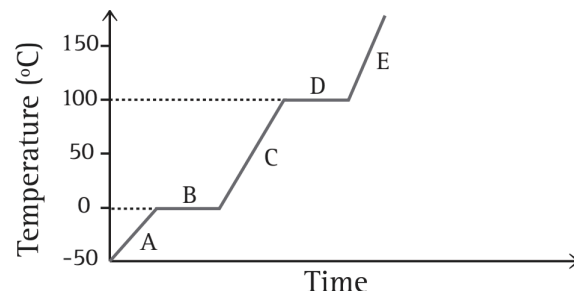
18. What is the correct formula for lithium oxide?

- a. LiO
- b. LiO_2
- c. Li_2O
- d. Li_2O_2

19. What is the molarity of an HCl solution if 50.0 mL of the solution is neutralized by 15.5 mL of 0.800 M NaOH?

- a. 0.248 M
- b. 0.645 M
- c. 0.496 M
- d. 1.29 M

20. The heating curve below shows what happens as heat is added to a piece of ice at a constant rate. What is occurring during the portion of the heating curve labeled A?



- a. Ice is melting to form liquid water.
- b. Ice is becoming warmer but not melting.
- c. Liquid water is freezing to become ice.
- d. Liquid water is boiling to form steam.

21. You burn a 5.0 g wood fuel pellet that warms 250 mL of water in an insulated calorimeter. The temperature of the water increases by 20 °C. How many calories of energy are contained in the pellet?

- a. 418 cal c. 20900 cal
b. 5000 cal d. 25000 cal

Base your answers to 22 on the table below, which shows the heat of combustion for six substances.

Substance	Chemical Formula	ΔH (kJ/mol)	ΔH (kJ/g)
Methane	CH ₄	-890	-55.5
Methanol	CH ₄ O	-2220	-50.3
Hexane	C ₆ H ₁₄	-2870	-49.4
Acetic Acid	C ₂ H ₄ O ₂	-1370	-29.7
Glucose	C ₆ H ₁₂ O ₆	-2800	-15.5
Octane	C ₈ H ₁₈	-5470	-47.9

22. Which of these substances has the largest heat of combustion, measured in kJ/mol?

- a. methane, CH₄
b. methanol, CH₄O
c. hexane, C₆H₁₄
d. acetic acid, C₂H₄O₂

Use the table below to answer question 23.

Bond	C—H	C—C	O—H	C—O	C=O	O=O
Average bond energy (kJ/mol)	413	347	467	358	799	146

23. What is the total energy required to break all the bonds in 1 mol of butane, C₄H₁₀?

- a. 5171 kJ c. 4130 kJ
b. 1048 kJ d. 760 kJ

24. Why do noble gases rarely form chemical bonds with other elements?

- a. They have a high atomic mass.
b. They have incomplete electron shells.
c. They are highly reactive.
d. Their electron configuration is stable.

25. Which reaction shows an example of beta decay?

- a. ${}_{92}^{238}\text{U} \rightarrow {}_{90}^{234}\text{Th} + {}_2^4\text{He}$
b. ${}_{6}^{14}\text{C} \rightarrow {}_{7}^{14}\text{N} + {}_{-1}^0\text{e}$
c. ${}_{27}^{60}\text{Co} \rightarrow {}_{27}^{60}\text{Co} + {}_0^0\gamma$
d. ${}_{6}^{11}\text{C} \rightarrow {}_{5}^{11}\text{B} + {}_{+1}^0\text{e}$

26. What is the piece of lab equipment shown on the right?

- a. Flask
b. Erlenmeyer Flask
c. Graduated cylinder
d. Pipette



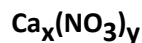
27. The mass of a rock is 625 g, and its density is 25 g/mL. What is the volume of the rock, in the appropriate units?

- a. 25 g c. 25 mL
b. 15625 g d. 15625 mL

28. Calculate the pH of a substance that has an [H⁺] of 0.0323 M.

- a. 0.49 c. 2.85
b. 1.20 d. 1.49

29. Identify the values for x and y in the following ionic compound:



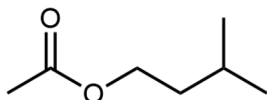
- a. $x = 1, y = 1$
- b. $x = 2, y = 1$
- c. $x = 1, y = 2$
- d. $x = 2, y = 2$

30. Name the following compound: $\text{Sn}(\text{C}_2\text{H}_3\text{O}_2)_2$

- a. Tin (I) acetate
- b. Tin (II) acetate
- c. Tin (III) acetate
- d. Tin (IV) acetate

31. What functional group exists on the molecule below?

- a. ether
- b. ester
- c. alcohol
- d. organic acid

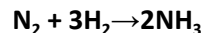


isopentyl acetate
(banana)

32. Which process is a chemical change?

- a. melting of ice
- b. synthesis of water
- c. subliming of ice
- d. deposition of water

33. In an experiment, a chemist reacts excess hydrogen gas with nitrogen gas to form ammonia (NH_3). The balanced equation is:



The **theoretical yield** of ammonia is **32.68 g**, but the **actual yield** obtained in the lab is **29.345 g**.

What is the percent yield of ammonia, rounded to the correct number of significant figures?

- a. 111.36%
- b. 111.4%
- c. 89.794%
- d. 89.79%

34. A student obtains the following four measurements of the boiling point of water:

99.8°C, 100.0°C, 100.1°C, 100.1°C

The standard deviation of these measurements would most likely be:

- a. Large, indicating low precision
- b. Small, indicating high precision
- c. Large, indicating high accuracy
- d. Small, indicating low accuracy

35. You burn a 5.0 g wood fuel pellet that warms 250 mL of water in an insulated calorimeter. The temperature of the water increases by 20 °C. How many calories of energy are contained in the pellet?

- a. 418 cal
- b. 5000 cal
- c. 20900 cal
- d. 2500 cal

Part B: Short Answer

Use your reference table and your knowledge of chemistry to answer the questions below. For no bonus points, draw a beaver next to question 36. [20]

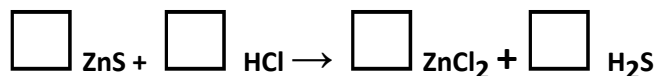
For the molecular formula for C_3H_9N :

36. Draw the structural formula.

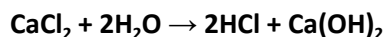
[1]

37. Balance the following double replacement equation:

[1]



38. Calcium chloride reacts with water to form hydrochloric acid and calcium hydroxide, as shown below. How many grams of hydrochloric acid will be produced from 7.00 g of water? [3]



39. How many milliliters of 0.45 M HCl must be added to 25.0 mL of 1.2 M KOH to make a neutral solution?

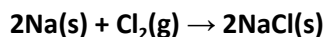
[1]

40. Complete the table below.

Formula	Structural Diagram [1] (Show all non bonding electrons.)	Ball & Stick Model [1]	Name of Shape [1]
CCl_4			

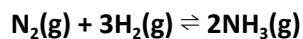
41. Is the molecule above polar? Explain your reasoning. [1]

Sodium chloride can be produced by the reaction of sodium metal with chlorine gas.



42. If you started with 161 g of Na and 150 g of Cl_2 , which reactant is limiting? [3]

43. A chemical reaction is at equilibrium in a closed container: [1]



A student wants to increase the *rate* at which the reaction reaches equilibrium. What is **one** factor the student can change to increase the rate of this reaction?

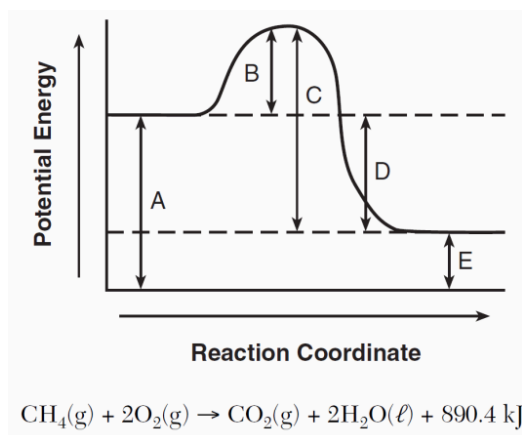
44. Draw a bohr diagram for an atom of Neon below.

[1]

45. A 135. g sample of water at 10°C is heated to steam at 105°C. How much heat does it absorb? [3]

$q = mC\Delta T$	$q = mL$
$c_{\text{water}} = 4.18 \text{ J/g}\cdot^{\circ}\text{C}$	$L_{\text{freezing/melting}} = 334 \text{ J/g}$
$c_{\text{ice}} = 1.99 \text{ J/g}\cdot^{\circ}\text{C}$	$L_{\text{condensing/vaporization}} = 2260 \text{ J/g}$
$c_{\text{vapor}} = 2.09 \text{ J/g}\cdot^{\circ}\text{C}$	

Base your answers to questions 46 and 47 on the information below. The chemical reaction between methane and oxygen is represented by the potential energy diagram and balanced equation below.



46. Explain, in terms of collision theory, why a lower concentration of oxygen gas decreases the rate of this reaction.

[1]

47. Which potential energy interval in the diagram represents the activation energy of the forward reaction?[1]

SCRAP - submit with your test